ACM-ICPC Team Reference Document

Skill++ (Gogolev, Drachev, Shafikov)

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1 General

1.1 C++ Template

```
#pragma GCC optimize("O3,unroll-loops")
#define _CRT_SECURE_NO_WARNINGS
#define ln '\n'
#define All(x) (x).begin(), (x).end()
#define Allr(x) (x).rbegin(), (x).rend()
#define Allf(x) x.begin() + 1, x.end()
#define fi(a, b) for (auto i = (a); i <= (b); i++)
#define fj(a, b) for (auto j = (a); j <= (b); j++)
#define fo(a, b) for (auto o = (a); o <= (b); o++)
#define fq(a, b) for (auto q = (a); q <= (b); q++)
#define fe(a, b) for (auto e = (a); e <= (b); e++)
#define fw(a, b) for (auto w = (a); w <= (b); w++)
#define fdi(a, b) for (auto i = (a); i >= (b); i--)
#define fdj(a, b) for (auto j = (a); j >= (b); j--)
#define fdo(a, b) for (auto o = (a); o >= (b); o--)
#define fdq(a, b) for (auto q = (a); q >= (b); q--)
#define fde(a, b) for (auto e = (a); e >= (b); e--)
#define fdw(a, b) for (auto w = (a); w >= (b); w--)
#define fx(A) for (auto &x : (A))
#define fy(A) for (auto &y : (A))
#define fz(A) for (auto &z : (A))
#define pb push_back
#define mp make_pair
#define ins insert
#define sz(x) (int)x.size()
#include <string>
#include <vector>
#include <list>
#include <map>
#include <set>
#include <deque>
#include <queue>
#include <stack>
#include <bitset>
#include <algorithm>
#include <sstream>
#include <iostream>
#include <iomanip>
#include <cstdio>
#include <cstdlib>
#include <ctime>
#include <cstring>
#include <cmath>
#include <cstdarg>
#include <cassert>
#include <ctime>
#include <tuple>
#include <unordered_set>
#include <unordered_map>
#include <random>
#include <chrono>
#include <complex>
using namespace std;
#ifdef _DEBUG
#define LOCAL
#endif
mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch
    ().count());
typedef long long ll;
typedef unsigned long long ull;
typedef long double ld;
typedef vector<bool> vb;
typedef vector<int> vi;
typedef vector<ll> vll;
typedef vector<ld> vd;
typedef vector<char> vc;
typedef vector<string> vs;
typedef vector<vb> vvb;
typedef vector<vi> vvi;
typedef vector<vll> vvll;
typedef vector<vc> vvc;
typedef pair<int, int> pii;
typedef pair<ll, ll> pll;
template<typename A, typename B>
ostream& operator<<(ostream& os, pair<A, B> p) {
    os << "(" << p.first << ", " << p.second << ")";
    return os;
}
template<typename T>
ostream& operator<<(ostream& os, vector<T> v) {
    fi(0, sz(v) - 1) {
        os << v[i] << " ";
    }
    return os;
}
template<typename T>
ostream& operator<<(ostream& os, set<T> t) {
    for (auto z : t) {
        os << z << " ";
    }
    return os;
}
template<typename T1, typename T2>
ostream& operator<<(ostream& os, map<T1, T2> t) {
    cerr << endl;
    for (auto z : t) {
        os << "\t" << z.first << " -> " << z.second << endl;
    }
    return os;
}
#ifdef LOCAL
```

```
#define dbg(x) {cerr << __LINE__ << "\t" << #x << ":␣"
      << (x) << endl;}
#else
#define dbg(x) {}
#endif
#ifdef LOCAL
#define ass(x) if (!(x)) { cerr << __LINE__ << "\tassertion␣
      failed:␣" << #x << endl, abort(); }
#else
#define ass(x) assert(x)
#endif
void solve() {

}
int main()
{
    ios::sync_with_stdio(0);
    cin.tie(0);
    cout.tie(0);
    cout << setprecision(20);
    #ifdef LOCAL
        freopen("input.txt", "r", stdin);
        freopen("output.txt", "w", stdout);
    #endif
    solve();
}
```

1.2 Compilation

```
#far association
g++ --std=c++17 !.! -DLOCAL -O2 -Wall -Wl,--stack
      =67108864 -o !.exe
```

1.3 Check

```
:start
gen.exe
sol.exe
copy output.txt answer.txt
NAME.exe
fc output.txt answer.txt
if ERRORLEVEL 1 goto end
goto start
:end
```

1.4 How to solve problems

1. Math formulation
2. Using parameters
3. Go to theory
4. Way od solutions:
   (a) fix parameter
   (b) cases
   (c) see on answer
   (d) induction
   (e) hypothesis
   (f) finding of
   (g) simplification
   (h) reduction to other problem
   (i) reduction to classic problem
   (j) reduction of classic problem to our
   (k) modeling
   (l) generalization
   (m) contrexamples
   (n) full random
   (o) invariant

2 Math

2.1 Simpson

```
double a = 0, b = 1;
const int n = 100000000;
double s = 0;
double h = (b - a) / n;
fi(0, n) {
        double x = a + h * i;
        s += fun(x) * ((i==0 || i==n) ? 1 : ((i&1)==0) ? 2 :4);
}
s *= h / 3;
```

2.2 Prime

```
ll fun1(ll a, ll n) {
        ll t = n - 1;
        int q = 0;
        while(t % 2 == 0) {
                q++;
                t /= 2;
        }
        ll v = pown(a, t, n);
        fi(1, q) {
                ll w = mult(v, v, n);
                if(v != 1 && v != n -1 && w == 1)return true;
                v = w;
        }
        if(v > 1)return true;
        return false;
}

bool is_prime(ll n) {
        if(n < 2)return false;
        if(n == 2)return true;
        if(n % 2 == 0)return false;
        fi(1, 10) {
                ll a = next(1, n - 1);
                if(fun1(a, n))return false;
        }
        return true;
}
```

2.3 Modulo

```
#define mod 1'000'000'007
int add(int a, int b){
        return (a + b) % mod;
}
int mult(int a, int b){
        return (ll)a * b % mod;
}
int pown(int a, ll n){
        if(n == 0)return 1;
        if(n % 2 == 0){
                int x = pown(a, n / 2);
                return mult(x, x);
        }
        return mult(a, pown(a, n - 1));
}
int obr(int a){
        return pown(a, mod - 2);
}
```

2.4 FFT

```
using cd = complex<double>;
const double PI = acos(-1);
void fft(vector<cd> & a, bool invert) {
    int n = a.size();

    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;

        if (i < j)
```

```
            swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len <<= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i+j], v = a[i+j+len/2] * w;
                a[i+j] = u + v;
                a[i+j+len/2] = u - v;
                w *= wlen;
            }
        }
    }

    if (invert) {
        for (cd & x : a)
            x /= n;
    }
}
vi multiply(vi const& a, vi const& b) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1;
    while (n < sz(a) + sz(b) ) {
        n <<= 1;
    }
    fa.resize(n);
    fb.resize(n);

    fft(fa, false);
    fft(fb, false);
    for (int i = 0; i < n; i++)
        fa[i] *= fb[i];
    fft(fa, true);

    vi result(n);
    for (int i = 0; i < n; i++)
        result[i] = round(fa[i].real());
    while(result.back() == 0)result.pop_back();
    return result;
}
```

2.5 NTT

```
pii fun(int lg) {
        int n = (next(0, MAX / (1 << lg) / 2) * 2 + 1) * (1 <<
                lg) + 1;
        while(!is_prime(n)) {
                n = (next(0, MAX / (1 << lg) / 2) * 2 + 1) * (1
                        << lg) + 1;
        }
        vi d;
        fj(1, n - 1) {
                if(j * j > n - 1)break;
                if( (n - 1) % j == 0) {
                        d.pb(j);
                        if(j * j != n - 1)d.pb((n - 1) / j);
                }
        }
        sort(All(d));
        d.pop_back();

        fj(2, n - 1) {
                bool f = 1;
                fx(d) {
                        if(pown(j, x, n) == 1) {
                                f = 0;
                                break;
                        }
                }
                if(f)return mp(n, j);
        }
        return mp(n, -1);
}
int LG = 19;
int MOD, G;
void fft(vi & a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;
        if (i < j)
            swap(a[i], a[j]);
    }
        for (int len = 2; len <= n; len <<= 1) {

        int wlen = G;
        if(invert)wlen = dv(1, wlen, MOD);
        for(int i = len;i < (1 << LG);i <<= 1) {
                wlen = mult(wlen, wlen, MOD);
        }

        for (int i = 0; i < n; i += len) {
            int w = 1;
            for (int j = 0; j < len / 2; j++) {
                int u = a[i+j];
                int v = mult(a[i+j+len/2], w, MOD);
                a[i+j] = add(u, v, MOD);
                a[i+j+len/2] = sub(u, v, MOD);
                w = mult(w, wlen, MOD);
            }
        }
    }
    if (invert) {
        int rn = dv(1, n, MOD);
        for (int & x : a)
            x = mult(x, rn, MOD);
    }
}
vi multiply(vi const& a, vi const& b) {
    vi fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1;
    while (n < sz(a) + sz(b) ) {
        n <<= 1;
    }

    fa.resize(n);
    fb.resize(n);
        fft(fa, false);
    fft(fb, false);
    for (int i = 0; i < n; i++)
        fa[i] = mult(fa[i], fb[i], MOD);
    fft(fa, true);
    while(fa.back() == 0)fa.pop_back();
    return fa;
}
auto T = fun(LG);
MOD = T.first;
G = T.second;
G = pown(G, (MOD - 1) / (1 << LG), MOD);
```

2.6 Gcd

```
int gcd(int a, int b, int& x, int& y) {
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    int x1, y1;
    int d = gcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}
bool find_any_solution(int a, int b, int c, int &x0, int &y0, int
      &g) {
    g = gcd(abs(a), abs(b), x0, y0);
    if (c % g) {
        return false;
    }

    x0 *= c / g;
    y0 *= c / g;
    if (a < 0) x0 = -x0;
    if (b < 0) y0 = -y0;
    return true;
}
```

2.7 Hungarian

```
vector<int> u (n+1), v (m+1), p (m+1), way (m+1);
for (int i=1; i<=n; ++i) {
    p[0] = i;
```

```
    int j0 = 0;
    vector<int> minv (m+1, INF);
    vector<bool> used (m+1, false);
    do {
        used[j0] = true;
        int i0 = p[j0], delta = INF, j1;
        for (int j=1; j<=m; ++j)
            if (!used[j]) {
                int cur = A[i0][j]-u[i0]-v[j];
                if (cur < minv[j])
                    minv[j] = cur, way[j] = j0;
                if (minv[j] < delta)
                    delta = minv[j], j1 = j;
            }
        for (int j=0; j<=m; ++j)
            if (used[j])
                u[p[j]] += delta, v[j] -= delta;
            else
                minv[j] -= delta;
        j0 = j1;
    } while (p[j0] != 0);
    do {
        int j1 = way[j0];
        p[j0] = p[j1];
        j0 = j1;
    } while (j0);
}
#get answer
vector<int> ans (n+1);
for (int j=1; j<=m; ++j)
    ans[p[j]] = j;
```

3 Strings

3.1 Suffix Array

```
struct Suff_arr{
  char LW = 3;
  int A = 256;
  string s0;
  string s1;
  vi arr;
  vi cost;
  vi rcost;
  vector<vector<pair<pair<int,int>,int > > > e1;
  vector<vector<pair<pair<int,int>,int > > > e2;
  vector<pair<pair<int,int>,int > > t;
  void Sort(vector<pair<pair<int,int>,int > >& t){
    fi(0,sz(t) - 1){
      e1[t[i].first.second].pb(t[i]);
    }
    fi(0,sz(e1) - 1){
      fx(e1[i]){
        e2[x.first.first].pb(x);
      }
      e1[i].clear();
    }
    t.clear();
    fi(0,sz(e2) - 1){
      fx(e2[i]){
        t.pb(x);
      }
      e2[i].clear();
    }

  }
  void init(string s){
    s0 = s;
    s1 = s;
    s1 += LW;
    fi(0,sz(s0) - 1){
      if( (sz(s1) & (sz(s1) - 1)) == 0)break;
      s1 += s0[i];
    }
    vvi e(A + 1);
    fi(0,sz(s1) -1){
      e[s1[i]].pb(i);
    }
    fi(0,A){
      fx(e[i]){
        arr.pb(x);
      }
    }
    cost = vi(sz(arr));
    rcost = vi(sz(arr));
    fi(1,sz(arr) - 1){
      cost[i] = cost[i-1];
      if(s1[arr[i]] != s1[arr[i-1]]){
        cost[i]++;
      }
    }
    int SIZ = 2;
    e1.resize(sz(arr));
    e2.resize(sz(arr));
    while(SIZ <= sz(arr)){
      fi(0,sz(arr) - 1){
        rcost[arr[i]] = cost[i];
      }
      t.clear();
      fi(0,sz(arr) - 1){
        t.pb( mp( mp(rcost[arr[i]], rcost[ (arr[i] + SIZ / 2)%sz(
            arr)]), arr[i]) );
      }
      Sort(t);
      cost[0] = 0;
      arr[0] = t[0].second;
      fi(1,sz(arr) - 1){
        arr[i] = t[i].second;
        cost[i] = cost[i- 1];
        if(t[i].first != t[i - 1].first)cost[i]++;
      }
      SIZ *= 2;
    }

    vi arr2;
    fx(arr){
      if(x < sz(s0)){
        arr2.pb(x);
      }
    }
    arr = arr2;
  }
  vi LCP;
  void init_lcp(){
    vi rarr(sz(arr));
    fi(0,sz(arr) - 1){
      rarr[arr[i]] = i;
    }
    LCP = vi(sz(arr));
    int len = 0;
    fi(0,sz(arr) - 1){
      if(rarr[i] == sz(arr) - 1){
        len = 0;
        continue;
      }
      len = max(0,len - 1);
      int x = i;
      int y = arr[rarr[i] + 1];
      while(x + len < sz(s0) && y + len < sz(s0) && s0[x + len
          ] == s0[y + len]){
        len++;
      }
      LCP[rarr[i]] = len;

    }

  }
};
```

3.2 Suffix Automaton

```
struct state {
        int len, link;
        map<char, int> next;
};
const int MAXLEN = 1'000'000;
state st[MAXLEN*2];
int siz, last;
void sa_init() {
        siz = last = 0;
        st[0].len = 0;
        st[0].link = -1;
        siz++;
}
void sa_extend (char c) {
        int cur = siz++;
        st[cur].len = st[last].len + 1;
        int p;
        for (p = last; p != -1 && !st[p].next.count(c); p = st[p].
            link) {
```

```
                st[p].next[c] = cur;
    }
        if (p == -1) {
                st[cur].link = 0;
        } else {
                int q = st[p].next[c];
                if (st[p].len + 1 == st[q].len) {
                        st[cur].link = q;
                } else {
                        int clone = siz++;
                        st[clone].len = st[p].len + 1;
                        st[clone].next = st[q].next;
                        st[clone].link = st[q].link;
                        for (; p != -1 && st[p].next[c] == q; p =
                                st[p].link)
                                  st[p].next[c] = clone;
                        st[q].link = st[cur].link = clone;
                }
        }
        last = cur;
}
```

3.3 Prefix function

```
vi prefix_function(string s) {
        int n = sz(s);
        vi pi(n);
        fi(1, n - 1) {
                int j = pi[i - 1];
                while (j > 0 && s[i] != s[j]) {
                        j = pi[j - 1];
        }
                if (s[i] == s[j]) j++;
                pi[i] = j;
        }
        return pi;
}
```

3.4 Z function

```
vi z_function (string s) {
        int n = sz(s);
        vi z (n);
        for (int i = 1, l = 0, r = 0; i < n; i++) {
                if (i <= r) z[i] = min(r - i + 1, z[i - l]);
                while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
                        z[i]++;
        }
                if (i + z[i] - 1 > r) {
            l = i;
            r = i + z[i] - 1;
        }
        }
        return z;
}
```

3.5 Aho Korasik

```
struct Node {
        int leaf = -1;
        char c = '\0';
        Node* par = nullptr;
        map<char, Node*> nxt;
        Node* suf = nullptr;
        map<char, Node*> sufs;
        Node(char t, Node* p) {
                c = t;
                par = p;
        }
};
Node* root;
void add(string& s, int j) {
        Node* cur = root;
        fi(0, sz(s)- 1) {
                if(cur->nxt.count(s[i])) {
                        cur = cur->nxt[s[i]];
                } else {
                        Node* t = new Node(s[i],cur);
                        cur->nxt[s[i]] = t;
                        cur = t;
                }
        }
        cur->leaf = j;
}
Node* get_link(Node* a) {
        if(a->suf) return a->suf;
        if(a == root) {
                a->suf = root;
                return a->suf;
        }
        if(a->par == root) {
                return root;
        }
        char t = a->c;
        Node* cur = get_link(a->par);
        while(cur != root && !cur->nxt.count(t)) {
                cur = get_link(cur);
        }
        if(cur->nxt.count(t)) {
                cur = cur->nxt[t];
        }
        return a->suf = cur;
}
Node* go(Node* a,char c) {
        Node* cur = a;
        while(cur != root && !cur->nxt.count(c)) {
                cur = get_link(cur);
        }
        if(cur->nxt.count(c)) {
                cur = cur->nxt[c];
        }
        return cur;
}

string s;
ll n;
vector<string> a;
void init() {
        root = new Node('\0',nullptr);
        fi(1, n) {
                add(a[i], i);
        }
}
```

4 Geometry

4.1 Manhatan Spanning

```
// Returns a list of edges in the format (weight, u, v).
// Passing this list to Kruskal algorithm will give the
      Manhattan MST.
vector<tuple<long long, int, int>> manhattan_mst_edges(
      vector<point> ps) {
    vector<int> ids(ps.size());
    iota(ids.begin(), ids.end(), 0);
    vector<tuple<long long, int, int>> edges;
    for (int rot = 0; rot < 4; rot++) { // for every rotation
        sort(ids.begin(), ids.end(), [&](int i, int j){
            return (ps[i].x + ps[i].y) < (ps[j].x + ps[j].y);
        });
        map<int, int, greater<int>> active; // (xs, id)
        for (auto i : ids) {
            for (auto it = active.lower_bound(ps[i].x); it !=
                  active.end();
            active.erase(it++)) {
                int j = it->second;
                if (ps[i].x - ps[i].y > ps[j].x - ps[j].y) break;
                assert(ps[i].x >= ps[j].x && ps[i].y >= ps[j].y);
                edges.push_back({(ps[i].x - ps[j].x) + (ps[i].y - ps[
                      j].y), i, j});
            }
            active[ps[i].x] = i;
        }
        for (auto &p : ps) { // rotate
            if (rot & 1) p.x *= -1;
            else swap(p.x, p.y);
        }
    }
    return edges;
}
```

5 Data Structures

5.1 Fenwick

```
// Linear
vll fen;
void add(int p, ll val) {
    for(int i = p; i <= n; i = (i | (i + 1))) {
        fen[i] += val;
    }
}
ll sum(int p) {
    ll res = 0;
    for(int i = p; i >= 0; i = (i & (i + 1)) - 1) {
        res += fen[i];
    }
    return res;
}

// Matrix
vvll fen;
void add(int x, int y, ll val) {
    for(int i = x; i <= n; i = (i | (i + 1))) {
        for(int j = y; j <= n; j = (j | (j + 1))) {
            fen[i][j] += val;
        }
    }
}
ll sum(int x, int y) {
    ll res = 0;
    for(int i = x; i >= 0; i = (i & (i + 1)) - 1) {
        for (int j = y; j >= 0; j = (j & (j + 1)) - 1) {
            res += fen[i][j];
        }
    }
    return res;
}
```

5.2 Long

```
struct Long {
    int base = 10000;
    int len = 4;
    vi num = {1, 0};
    int operator[](int x) const {
        if (x > num[0]) return 0;
        return num[x];
    }
    int& operator[](int x) {
        if (x > num[0]) {
            num.resize(x + 1, 0);
            num[0] = x;
        }
        return num[x];
    }
};
Long operator+(const Long& A, const Long& B) {
    Long res;

    int n = max(A[0], B[0]);
    fi(1, n) {
        res[i] = A[i] + B[i];
        if (i > 1) {
            res[i] += res[i - 1] / res.base;
            res[i - 1] %= res.base;
        }
    }
    while(res[res[0]] >= res.base) {
        int m = res[0];
        res[m + 1] = res[m] / res.base;
        res[m] %= res.base;
    }
    return res;
}
Long operator*(const Long& A, const int& B) {
    Long res;
    int n = A[0];
    fi(1, n) {
        res[i] = A[i] * B;
        if (i > 1) {
            res[i] += res[i - 1] / res.base;
            res[i - 1] %= res.base;
        }
    }
    while(res[res[0]] >= res.base) {
        int m = res[0];
        res[m + 1] = res[m] / res.base;
        res[m] %= res.base;
    }
    return res;
}
Long operator*(const Long& A, const Long& B) {
    Long res;
    fi(1, A[0]) {
        if (A[i] == 0) continue;
        fj(1, B[0]) {
            if (A[i] * B[j] == 0) continue;
            int x = i + j - 1;
            res[x] += A[i] * B[j];
            while(x > 1 && res[x - 1] >= res.base) {
                res[x] += res[x - 1] / res.base;
                res[x - 1] %= res.base;
            }
        }
        while(res[res[0]] >= res.base) {
            int m = res[0];
            res[m + 1] += res[m] / res.base;
            res[m] %= res.base;
        }
    }
    return res;
}
pair<Long, int> operator/(const Long& A, const int B) {
    Long res;
    ll x = 1;
    ll d = 0, m = 0;
    fdi(A[0], 1) {
        ll val = m * res.base + A[i];
        d = val / B;
        m = val % B;
        if (x == 1 && d == 0) continue;
        res[x] = d;
        x++;
    }
    reverse(Allf(res.num));
    return mp(res, m);
}
void print(const Long& A) {
    printf("%d", A[A[0]]);
    fdi(A[0] - 1, 1) {
        printf("%04d", A[i]);
    }
    printf(ln);
}
Long str_to_Long(const string& s) {
    int n = sz(s);
    Long res;
    int x = 1;
    for(int i = n - 1; i >= 0; i -= 4) {
        int val = 0;
        for(int j = max(0, i - 3); j <= i; j++) {
            val = val * 10 + (s[j] - '0');
        }
        res[x] = val;
        x++;
    }
    return res;
}
Long int_to_Long(int num) {
    Long res;
    int x = 1;
    while(num) {
        int val = num % res.base;
        num /= res.base;
        res[x] = val;
        x++;
    }
    return res;
}
```

5.3 Persistent treap

```
int next(int l,int r){

        return l + rnd()%(r - l + 1);
}
const int MAX = 1e7;

int siz = 1;
vi Left(MAX,-1),Right(MAX,-1),Siz(MAX,0),Val(MAX,0);
```

```
void upd(int v){
        if(v == -1)return;
        Siz[v] = 1;
        if(Left[v] != -1)Siz[v] += Siz[Left[v]];
        if(Right[v] != -1)Siz[v] += Siz[Right[v]];
}

bool INIT;
int cop(int v){
        if(INIT)return v;
        int x = siz;
        siz++;
        Val[x] = Val[v];
        Left[x] = Left[v];
        Right[x] = Right[v];
        Siz[x] = Siz[v];
        return x;
}

int merg(int L,int R){
        if(L == -1)return R;
        if(R == -1)return L;

        if(next( 1, (Siz[L] + Siz[R])*100 ) <= Siz[L] * 100){
                L = cop(L);
                Right[L] = merg(Right[L], R);
                upd(L);
                return L;
        }else{
                R = cop(R);
                Left[R] = merg(L,Left[R]);
                upd(R);
                return R;
        }
}
pair<int,int > split(int v,int k){
        if(Siz[v] <= k){
                return mp(v,-1);
        }
        if(k == 0){
                return mp(-1,v);
        }
        v = cop(v);
        int siz = 0;
        if(Left[v] != -1)siz = Siz[Left[v]];
        if(siz >= k){
                auto t = split(Left[v],k);
                Left[v] = t.second;
                upd(v);
                return mp(t.first,v);
        }else{
                auto t = split(Right[v], k - siz - 1);
                Right[v] = t.first;
                upd(v);
                return mp(v,t.second);
        }

}

int root = -1;
void seg_copy(int i,int j,int k){
        auto t = split(root,j - 1);
        auto g = split(root,i - 1);
        auto h = split(g.second,k);
        auto f = split(t.second,k);
        root = merg(t.first,merg(h.first,f.second));
}

int n,m;
vi cnt,from,to;
vi res;

void solve() {
        INIT = true;
        fi(1, n){
                int v = siz++;
                upd(v);
                Val[v] = i;

                root = merg(root,v);
        }
        INIT = false;
}
```

5.4 Segment updates treap

```
const ll E = 0;

struct Node {
    ll val;
    ll buf;
    ll min_val;
    ll cnt;
    ll y;
    Node *l;
    Node *r;
};

void push(Node* a) {
        if(!a) return;
        if(a->buf == E) return;
    a->val += a->buf;
    a->min_val += a->buf;

    if(a->l) {
                a->l->buf += a->buf;
        // a->l->min_val += a->buf;
        }
        if(a->r) {
                a->r->buf += a->buf;
        // a->r->min_val += a->buf;
        }
        a->buf = E;
}
void upd(Node *a) {
    a->cnt = 1;
    a->min_val = a->val;
    if (a->l) {
        a->min_val = min(a->min_val, a->l->min_val + a->l
            ->buf);
        a->cnt += a->l->cnt;
    }
    if (a->r) {
        a->min_val = min(a->min_val, a->r->min_val + a->r
            ->buf);
        a->cnt += a->r->cnt;
    }
}
Node* new_node(ll val) {
        Node* x = new Node;
        x->y = rnd();
        x->val = val;
        x->buf = E;
        x->cnt = 1;
        x->l = x->r = 0;
        upd(x);
        return x;
}
Node *merge(Node *a, Node *b) {
    push(a);
    push(b);
    if (!a)
        return b;
    if (!b)
        return a;
    if (a->y > b->y) {
        a->r = merge(a->r, b);
        upd(a);
        return a;
    }
    b->l = merge(a, b->l);
    upd(b);
    return b;
}
pair<Node *, Node *> split(Node *a, ll q) {
    push(a);
    if (!a)
        return {0, 0};

    ll left = 0;
    if (a->l)
        left += a->l->cnt;
    if (left >= q) {
        auto [l, r] = split(a->l, q);
        a->l = r;
        upd(a);
        return {l, a};
    }
    auto [l, r] = split(a->r, q - left - 1);
    a->r = l;
    upd(a);
    return {a, r};
```

```
}
```

5.5 Treap

```
struct Node {
    ll val;
    ll cnt;
    ll y;
    Node *l;
    Node *r;
};
void upd(Node *a) {
    a->cnt = 1;
    if (a->l)
        a->cnt += a->l->cnt;
    if (a->r)
        a->cnt += a->r->cnt;
}
Node *merge(Node *a, Node *b) {
    if (!a)
        return b;
    if (!b)
        return a;
    if (a->y > b->y) {
        a->r = merge(a->r, b);
        upd(a);
        return a;
    }
    b->l = merge(a, b->l);
    upd(b);
    return b;
}
pair<Node *, Node *> split(Node *a, ll q) {
    if (!a)
        return {0, 0};
    ll left = 0;
    if (a->l)
        left += a->l->cnt;
    if (left >= q) {
        auto [l, r] = split(a->l, q);
        a->l = r;
        upd(a);
        return {l, a};
    }
    auto [l, r] = split(a->r, q - left - 1);
    a->r = l;
    upd(a);
    return {a, r};
}
```

6 DP

6.1 Knuth

```
int solve() {
    int N;
    ... // read N and input
    int dp[N][N], opt[N][N];

    auto C = [&](int i, int j) {
        ... // Implement cost function C.
    };

    for (int i = 0; i < N; i++) {
        opt[i][i] = i;
        ... // Initialize dp[i][i] according to the problem
    }

    for (int i = N-2; i >= 0; i--) {
        for (int j = i+1; j < N; j++) {
            int mn = INT_MAX;
            int cost = C(i, j);
            for (int k = opt[i][j-1]; k <= min(j-1, opt[i+1][j]); k
                ++) {
                if (mn >= dp[i][k] + dp[k+1][j] + cost) {
                    opt[i][j] = k;
                    mn = dp[i][k] + dp[k+1][j] + cost;
                }
            }
            dp[i][j] = mn;
        }
    }

    return dp[0][N-1];
}
```

6.2 Convex hull trick

```
struct Line {
    mutable ll k, m, p;
    bool operator<(const Line &o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};
struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = INF;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y) {
        if (y == end())
            return x->p = inf, 0;
        if (x->k == y->k)
            x->p = x->m > y->m ? inf : -inf;
        else
            x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z))
            z = erase(z);
        if (x != begin() && isect(--x, y))
            isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};
```